MTU_Value*Service*
Technical Documentation

Diesel Engine
12 V 4000 R33
Application Group 2A

Maintenance Schedule
MS50007/00E

1 Maintenance Schedule

1.1 Preface

MTU maintenance concept

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. The time intervals at which the maintenance work is to be carried out and the relevant checks and tasks involved are average values based on operational experience and are therefore to be regarded as guidelines only. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.
QL2: Component exchange (corrective only).
QL3: Maintenance work which requires partial disassembly of the engine.
QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:
  The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).
- Battery:
  Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not installed on your engine; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application Group

2A	Continuous operation, unrestricted

TIM ID: 0000000002 - 001

Load profile

Load factor	100%	60%	50%	30%	<15%
Corresponding operating time	0,5%	4,5%	10%	20%	65%

ANNUALHOURS: Maintenance schedule applicable to maximum annual engine hours less than 3000.

For engine hours in excess of 3000 hours per year, adjust accordingly.

1.2 Maintenance schedule matrix

0-21.000 Operating hours

Item	Limit	Operating hours [h]																					
		Daily	1.000	2.000	3.000	4.000	5.000	6.000	7.000	8.000	9.000	10.000	11.000	12.000	13.000	14.000	15.000	16.000	17.000	18.000	19.000	20.000	21.000
Engine operation	-	X																					
Air filters	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Oil indicator filter	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Valve gear	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Fuel filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Centrifugal oil filter	-			X		X		X		X		X		X		X		X		X		X	
Coolant filters	2 a				X			X			X			X			X			X			X
Fuel injectors	-							X						X						X			
Combustion chambers	-							X						X						X			
Air filters	3 a							X						X						X			
Turbochargers	-											X										X	
Component maintenance	-											X										X	
Fuel delivery pump	-											X										X	
HP fuel pump	-																X						
Cylinder heads	-																X						
Crankcase breathers	-																						
Automatic engine oil filter	-																						
Extended component maintenance	10 a																						
w = weeks m = months a = years																							

22.000-30.000 Operating hours

Item	Limit	Operating hours [h]																					
		22.000	23.000	24.000	25.000	26.000	27.000	28.000	29.000	30.000													
Engine operation	-																						
Air filters	-	X	X	X	X	X	X	X	X	X													
Oil indicator filter	-	X	X	X	X	X	X	X	X	X													
Valve gear	-	X	X	X	X	X	X	X	X	X													
Fuel filter	2 a	X	X	X	X	X	X	X	X	X													
Centrifugal oil filter	-	X		X		X		X		X													
Coolant filters	2 a			X			X			X													
Fuel injectors	-			X						X													
Combustion chambers	-			X						X													
Air filters	3 a			X						X													
Turbochargers	-									X													
Component maintenance	-									X													
Fuel delivery pump	-									X													
HP fuel pump	-									X													
Cylinder heads	-									X													
w = weeks m = months a = years																							

TIM ID: 0000000003 - 001

Item	Limit	Operating hours [h]																					
		22.000	23.000	24.000	25.000	26.000	27.000	28.000	29.000	30.000													
Crankcase breathers	-									X													
Automatic engine oil filter	-									X													
Extended component maintenance	10 a																						
w = weeks m = months a = years																							

1.3 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Task
QL1	Daily	-	Engine operation	Check engine oil level (→ Operating Instructions)	W0500
				Carry out visual inspection of engine for general condition and leaks (→ Operating Instructions)	W0501
				Inspect intercooler drain system (if fitted) (→ Operating Instructions)	W0502
				Inspect service indicator of air filter (→ Operating Instructions)	W0503
				Check relief bores of coolant pump(s) (→ Operating Instructions)	W0505
				Check for abnormal running noises, exhaust gas color and vibration (→ Operating Instructions)	W0506
				Drain off water and contamination from fuel prefilter (if fitted) (→ Operating Instructions)	W0507
				Check service indicator of fuel prefilter (if fitted) (→ Operating Instructions)	W0508
QL1	1000	-	Air filters	Clean air filter(s) Empty dust collector box (if fitted)	W1027
QL1	1000	-	Oil indicator filter	Check and clean oil indicator filter (→ Operating Instructions)	W1047
QL1	1000	-	Valve gear	Check valve clearance (→ Operating Instructions)	W1002
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert (→ Operating Instructions)	W1001
QL1	2000	-	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve (if fitted) (→ Operating Instructions)	W1009
QL1	3000	2 a	Coolant filters	Fit new coolant filters (→ Operating Instructions)	W1036
QL1	6000	-	Fuel injectors	Fit new fuel injectors (→ Operating Instructions)	W1006
QL1	6000	-	Combustion chambers	Inspect cylinder chambers using endoscope (→ Operating Instructions)	W1011
QL1	6000	3 a	Air filters	Fit new air filters (→ Operating Instructions)	W1005
QL3	10000	-	Turbochargers	Overhaul turbochargers	W1038
QL3	10000	-	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems (→ Instructions for Overhaul of Assemblies)	W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces (→ Instructions for Overhaul of Assemblies)	W2001
				Clean air ducting (→ Instructions for Overhaul of Assemblies)	W2002
				Clean intercooler and inspect for leakage (→ Instructions for Overhaul of Assemblies)	W2003
				Fit new high-pressure fuel sensor (→ Instructions for Overhaul of Assemblies)	W2004
				Inspect centrifugal oil filter for wear (if fitted) (→ Instructions for Overhaul of Assemblies)	W2009
				Overhaul starter (→ Instructions for Overhaul of Assemblies)	W2010
				Inspect vibration damper (→ Instructions for Overhaul of Assemblies)	W2011
				Clean engine coolant cooler and, if possible, inspect it for leaks (→ Instructions for Overhaul of Assemblies)	W2017
				Clean engine oil cooler and inspect it for leaks (→ Instructions for Overhaul of Assemblies)	W2018
				Disassemble Küsel coupling (if fitted). Check parts. Fit new friction rings and disks (→ Instructions for Overhaul of Assemblies)	W2019
				Check adjustment at central buffer of engine mounts (→ Instructions for Overhaul of Assemblies)	W2020
				Overhaul charge air coolant pump (→ Instructions for Overhaul of Assemblies)	W2070
				Fit new seals/sealing materials for all disassembled components (→ Instructions for Overhaul of Assemblies)	W2062
				Overhaul engine coolant pump (→ Instructions for Overhaul of Assemblies)	W2110

w = weeks
m = months
a = years

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Task
				Check drive of fuel delivery pump (→ Instructions for Overhaul of Assemblies)	W2131
				Check engine coolant thermostat and fit new thermal actuator (→ Instructions for Overhaul of Assemblies)	W2006
				Check charge-air coolant thermostat and fit new thermal actuator (→ Instructions for Overhaul of Assemblies)	W2007
QL3	10000	-	Fuel delivery pump	Fit new fuel delivery pump (→ Instructions for General Overhaul)	W1051
QL3	15000	-	HP fuel pump	Fit new high-pressure fuel pump (→ Instructions for Overhaul of Assemblies)	W1058
QL3	15000	-	Cylinder heads	Overhaul cylinder heads	W1134
QL3	30000	-	Crankcase breathers	Crankcase breathers: Fit new filters or filter inserts	W1046
QL3	30000	-	Automatic engine oil filter	Remove filter housing and clean. Fit new filter candles (→ Operating Instructions)	W1035
QL4	-	10 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required (→ Instructions for General Overhaul)	W3000
				Replace all elastomeric parts and seals with new ones (→ Instructions for General Overhaul)	W3001
				Fit new piston rings (→ Instructions for General Overhaul)	W3002
				Fit new conrod bearings (→ Instructions for General Overhaul)	W3003
				Fit new crankshaft bearings (→ Instructions for General Overhaul)	W3004
				Fit new cylinder liners (→ Instructions for General Overhaul)	W3005
				Fit new antifriction bearings for auxiliary PTOs (→ Instructions for General Overhaul)	W3006
				Fit new pressure relief valve in high-pressure fuel system (→ Instructions for General Overhaul)	W3021
				Fit new compressor wheels for turbochargers (→ Instructions for General Overhaul)	W3060
				Replace wiring harnesses	W3083
w = weeks m = months a = years					

TIM ID: 0000000004 - 001